

Product Requirement Document: Online Archery Gaming Website

1. Overview

Online Archery Gaming Website is a single-player, browser-based archery game built by Quivermark Studios as a self-contained MVP. The game places the player in the role of an archer who draws a bow, accounts for wind, and shoots 10 arrows at a 5-ring target across 3 difficulty levels, competing for a place on a local top 10 leaderboard. The primary business objective is to validate the core aim, draw, release, and score gameplay loop and to serve as a portfolio-grade showcase of HTML5 Canvas game development before committing resources to a full production title.

2. Project Context

Field	Detail
Game Type	Online archery game (browser based)
Category	Web Development
Platform	Web browser, desktop and mobile responsive
Minimum width	320px
Base resolution	1280x720 (desktop)
Tech stack	HTML5, CSS3, Vanilla JavaScript (Canvas API)
Target FPS	60
Deployment	Static files; runs by opening <code>index.html</code>

No server, no database, and no network call is required after the page loads. All persistence uses `localStorage`.

3. Scope of Work

S.No	Work Item	Description
1	Game engine	Canvas render loop targeting 60 FPS using <code>requestAnimationFrame</code>
2	Bow mechanic	Power bar charged by holding mouse or tap; release fires the arrow
3	Target and scoring	5 ring target; distance based hit detection; 0 to 10 points
4	Wind system	On screen wind indicator that alters arrow trajectory
5	Difficulty levels	3 levels with distinct, measurable parameters
6	Round logic	10 arrows per round; per difficulty countdown timer
7	Leaderboard	Top 10 scores in <code>localStorage</code> , persistent across refresh
8	Screens and navigation	4 screens, all reachable without page reload
9	UI theme	Medieval and nature theme using the fixed palette and fonts below
10	Documentation	<code>README.md</code> with setup and play instructions

4. Provided Materials

Material	Provided	Notes
<code>reference_image.png</code>	Yes	Visual reference for the game look and feel

Only `reference_image.png` is provided. Fonts and all other elements not listed above are chosen at the developer's discretion, provided they fit the medieval and nature theme and palette defined in Section 6.

4.1 Required Assets

All assets are sourced independently by the developer. All assets must be free for commercial use. No AI-generated assets are permitted.

Images

Asset	Format	Suggested Source
Archery target (5 rings)	PNG / SVG	OpenGameArt.org
Bow, idle state	PNG	OpenGameArt.org
Bow, drawn state	PNG	OpenGameArt.org
Arrow	PNG	OpenGameArt.org
Forest background (1280x720)	PNG	Kenney.nl
Wood panel texture (UI)	PNG	Kenney.nl
Wind direction icon	PNG / SVG	Flaticon (free tier)

Sounds

Sound	Format	Suggested Source
Bow draw	MP3 / WAV	Freesound.org
Arrow release	MP3 / WAV	Freesound.org
Bullseye hit	MP3 / WAV	Freesound.org
Regular hit	MP3 / WAV	Freesound.org
Miss (whoosh)	MP3 / WAV	Freesound.org
Round complete	MP3 / WAV	Freesound.org

Asset Rules

- All assets must be licensed for free commercial use.
- Preferred formats: PNG for images, MP3 for audio.
- The submission includes an attribution.txt file inside the assets/ folder listing every asset source and license.
- Asset filenames must follow: ARCHERY_[asset-name].[ext]. Example: ARCHERY_bow-idle.png

5. The Product / Game

Spec	Value
Arrows per round	10
Target rings	5
Maximum round score	100 pts
Bow input	Hold mouse or tap to charge power bar; release to shoot
Power bar range	0% to 100%
Wind	On screen indicator; alters arrow horizontal trajectory
Difficulty levels	3 (Easy, Medium, Hard)
Leaderboard	Top 10 scores, localStorage
Initials input	3 characters
Render target	60 FPS via requestAnimationFrame

5.1 Scoring and Difficulty

Ring	Points	Color
Bullseye	10	Deep Red #8B1A1A
Gold	8	#8B6914
Red	6	Crimson Red #C0392B
Blue	4	Sky Blue #87CEEB
Black	2	#000000
Miss	0	n/a

Note: Crimson Red #C0392B is approved exclusively for the Red ring on the target. It does not apply to any UI element, screen, or component outside the target canvas.

Target Ring Radii (at 1280x720 baseline)

Ring	Radius (px)
Bullseye	20px
Gold	40px
Red	65px
Blue	95px
Black	130px

Radii scale proportionally when canvas is resized below 1280x720.

Level	Target Size	Wind	Wind Speed Range	Round Timer	Target Scale
Easy	Large	Slow	±1 to ±3 px per frame	60s	1.0x (outer ring radius 130px)
Medium	Medium	Moderate	±4 to ±6 px per frame	45s	0.7x (outer ring radius 91px)
Hard	Small	Strong, random	±7 to ±12 px per frame, randomized each arrow	30s	0.45x (outer ring radius 58px)

Hit detection uses the Euclidean distance from the arrow landing point to the target center, compared against each ring radius. The bullseye is the smallest ring and awards the highest value.

6. Design Direction

Theme: Medieval and nature. Flat shapes, wood and parchment surfaces, target archery focus. No gradients, no neon colors, no anime style, no drop shadows or glow.

Buttons use a consistent corner radius of 6px.

Purpose	Color Name	Hex
Background, primary	Forest Green	#2D5A27
UI panels, borders	Aged Wood	#8B6914
Text, labels	Cream Parchment	#F5E6C8
Bullseye, accents	Deep Red	#8B1A1A
Wind indicator, highlights	Sky Blue	#87CEEB

Only these 5 colors plus pure black (#000000) and pure white (FFFFFF) for contrast are permitted. Fonts are chosen at the developer's discretion to fit the medieval and nature theme. Text contrast must meet a minimum ratio of 4.5:1.

7. Deliverable Requirements

7.1 Home Screen

Element	Requirement
Title	Game title in a display font chosen by the developer to fit the theme
Play button	Starts a round at the selected difficulty
Leaderboard button	Opens the Leaderboard Screen
Difficulty selector	Selects Easy, Medium, or Hard before play

7.2 Game Screen

Element	Requirement
Canvas	Renders the bow, arrow, target, and wind
Score display	Running round score, updated within 100ms of each arrow landing
Arrow count	Arrows remaining, format Arrows: X / 10
Wind indicator	Direction and strength, drawn in Sky Blue #87CEEB
Timer	Countdown per difficulty (60s, 45s, 30s)
Power bar	Fills 0% to 100% while input is held; resets after each shot
Aim indicator	Set by the pointer position relative to the bow; updates every animation frame while aiming
Difficulty label	Active difficulty name shown on screen during play

Power bar behavior: The bar fills from 0% to 100% linearly in 2 seconds while input is held. It holds at 100% until the player releases. It does not cycle back to 0% and does not auto-fire. The bar resets to 0% immediately after each shot.

The power bar renders as a bar with a width of 200px at the 1280x720 desktop baseline.

7.3 Results Screen

Element	Requirement
Round score	Score for the completed round
Total score	Cumulative session total score
Initials input	3 character text entry
Play Again button	Starts a new round
Home button	Returns to the Home Screen

7.4 Leaderboard Screen

Element	Requirement
Scores table	Top 10 scores, sorted descending
Entry fields	Initials (3 chars), score, difficulty
Persistence	Loaded from <code>localStorage</code> , survives page refresh
Back button	Returns to the Home Screen
Empty state	Message shown when no scores exist

7.5 Game Logic

Element	Requirement
Loop	<code>requestAnimationFrame</code> , 60 FPS target
Physics	Custom gravity, velocity, and wind offset in vanilla JS
Round	Ends after 10 arrows or when the timer reaches 0
Navigation	All 4 screens reachable without page reload

Arrow physics baselines: Initial arrow velocity is proportional to the power bar percentage at release, with a maximum of 15px per frame at 100% power. Gravity applies a constant downward acceleration of 0.3px per frame squared. Wind offset is applied horizontally each frame using the wind speed range defined in Section 5.1 for the active difficulty level. These values are baselines; minor tuning for game feel is permitted provided scoring and hit detection remain consistent.

Each landed arrow remains visibly stuck in the target until the round ends, so the player can see their shot grouping.

8. Required Deliverables

S.No	Item	Filename	Format	Notes
1	Main entry point	<code>index.html</code>	.html	Opens and runs the game
2	Styling	<code>style.css</code>	.css	All styling; theme and palette applied
3	Game logic	<code>game.js</code>	.js	All gameplay, scoring, leaderboard
4	Assets	<code>assets/</code>	folder	All images and sounds used; images under <code>assets/images/</code> , sounds under <code>assets/sounds/</code>
5	Documentation	<code>README.md</code>	.md	Setup and play instructions

9. Conventions

Convention	Rule
File naming	<code>ARCHERY_[item_name]_v1_[YYYYMMDD].[ext]</code>
Example	<code>ARCHERY_game_logic_v1_20260614.js</code>
Case	lowercase for the item name
Spaces	Not allowed in filenames
Exception	<code>index.html</code> keeps its exact name to open directly
Exception (core files)	<code>index.html</code> , <code>style.css</code> , <code>game.js</code> , and <code>README.md</code> keep their exact names. All other files including assets follow the <code>ARCHERY_</code> naming convention.
Storage key	Namespaced, for example <code>archery_leaderboard</code>
Code style	Commented, sensibly organized, vanilla JS only

Tool Requirements

Item	Rule
Required tools	A plain text or code editor and a web browser for testing
Acceptable alternatives	Any code editor; any static local server for testing
Forbidden	Paid plugins or themes, no-code or site-builder exports, and framework CLIs that inject a build dependency

10. Quality Standards

Standard	Requirement
Frame rate	60 FPS on latest Chrome (desktop)

Console	No errors on launch
Responsiveness	Fully playable down to 320px width
Persistence	<code>localStorage</code> leaderboard survives page refresh
Navigation	All 4 screens reachable without page reload
Offline	Runs after first load with no network calls
Browser support	Latest stable Chrome, Edge, Firefox, and Safari (desktop and mobile)
High-DPI rendering	Account for devicePixelRatio so rendering is crisp on high-DPI screens
Source files	Human-readable, unminified HTML, CSS, and JS; a minified copy is optional but the readable source is mandatory
Aspect ratio	1280x720 baseline, 16:9 preserved on scale

Screenshots and Export

Requirement	Detail
Minimum screenshot size	1280 px wide
Color mode	sRGB
Image quality	PNG preferred; if JPG, 85% quality or higher
Required screenshots	Start/difficulty screen, an active round, and the leaderboard
Readability	All UI text remains fully readable; no cropping of the score, arrow count, or buttons

11. Acceptance Criteria

- Game renders on an HTML5 canvas at a 1280x720 baseline and scales down to 320px width.
- Game loop runs via `requestAnimationFrame` at 60 FPS on latest Chrome.
- Bow charges by holding mouse or tap; power bar fills 0% to 100% and releases on input release.
- Target has exactly 5 rings scoring 10, 8, 6, 4, 2; a miss scores 0; maximum round score is 100.
- Bullseye is the smallest ring, colored Deep Red `#8B1A1A`, and awards 10 points.
- Each round is exactly 10 arrows.
- 3 difficulty levels exist: Easy (large target, slow wind, 60s), Medium (medium target, moderate wind, 45s), Hard (small target, strong random wind, 30s).
- Wind indicator is shown on screen and measurably alters arrow trajectory.
- Round ends after 10 arrows or when the timer reaches 0.
- Leaderboard stores the top 10 scores in `localStorage`, sorted descending, and persists across page refresh.
- Results Screen accepts a 3 character initials entry and offers Play Again and Home buttons.
- All 4 screens (Home, Game, Results, Leaderboard) are reachable without a page reload.
- Palette uses only the 5 defined hex values plus black and white; fonts are the developer's choice within the medieval and nature theme.
- No gradients, no neon colors, no anime style.
- No console errors on launch.
- All files follow the `ARCHERY_[item_name]_v1_[YYYYMMDD].[ext]` naming convention.

12. Scope Boundaries

In Scope

- Single player, client side archery game.
- Bow and power mechanic, wind system, 5 ring scoring, 3 difficulty levels.
- 10 arrow rounds with per difficulty timers.
- Local top 10 leaderboard via `localStorage`.
- 4 screens with no reload navigation.
- Medieval and nature theme using the fixed palette and fonts.

Out of Scope

- Backend, server, or database.
- Authentication or login.
- Paid libraries or plugins.
- Multiplayer.
- Unity, React, or any framework. Vanilla JS only.